8D report

Customer plant:	Bosch China	**Supplier:**	AVO Carbon Germany	**Date of claim:**	08.01.2025
Customer contact:	Linhua ZHU	**AVO contact:**	Gunther Luschtinetz		
Customer reference:	E-Mail 08.01.2025	**AVO reference:**	250101		

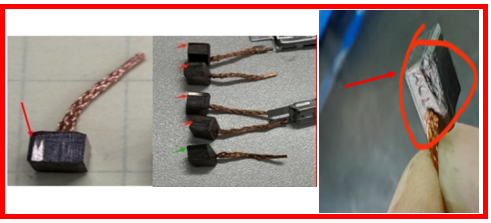
QUICK RESPONSE (D1-D3)
This document must be issued within 24 hours after claim notification

D 1. Description of the problem

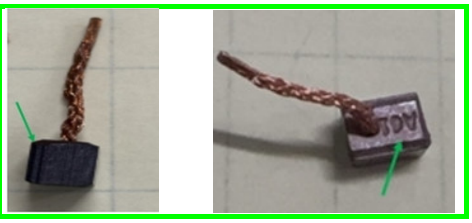
1.a What is the problem? Use 5W+2H to describe what is the current situation.

Problem description	1394320515 Surface damage and abnormal	What happened	1394320515 Surface damage and abnormal
		Why is it a problem	Affect the function
Incident type	Customer claim		
Part number AVO	1026644	When did it happen	08.01.2025
Part number Customer	1394320515	Who detected it	Line Operator
Batch number	281927	Where was it detected	Shenda assy. Line
Recurrence	Non-recurrent	How was it detected	Visually
Defect S/R	No	How many bad parts	100 parts (3,2%)
Standard process used	Yes		
When was the part produced?	Pressing machine => X12-11: 11.10.24-22.10.24 Grinding machine => SSS 18030: 24.10.24-25.10.24		

nOK part picture/ sketch



OK part picture/ sketch



1.b Problem solving team

Name	Position	Location	E-mail
Gunther Luschtinetz	Quality Management	AVO Germany	gunther.luschtinetz@avocarbon.com
Daniel Beil	Quality Management	AVO Germany	daniel.beil@avocarbon.com
Khaled Amri	Logistics	AVO Germany	
Martin Röhr	Production	AVO Germany	
Mustafa Yorulmaz	Production	AVO Germany	

1.c Tools to use

Yes	Ishikawa	Yes	5why's		LLC		Audit		Other

1.d First analyse of problem / potential root causes

Potential root cause(s):Defect cylinder, defect spindel

If you use our expressnumber, please use only economic express.

Samples of nok parts required in plant for analyse: Yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts?		Comments (compulsory, if yes)
Other family product	No	
Other line	No	
Other tools/work station	No	
Other	No	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	N° of involved parts	N° parts sorted	N° nok parts	Ppm	Date	Responsible
Parts at customer	---	---	100	---	08.01.2025	Linhua ZHU
Parts in transport	---	---	---	---	---	---
Parts at AVO Germany stock	---	---	---	---	---	Gunther Luschtinetz
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Containment actions	Responsible	Date planned	Date closed
Register the complaint + Quality alert	Gunther Luschtinetz	08.01.2025	08.01.2025
Check the production-, quality- and maintenance data	Gunther Luschtinetz	08.01.2025	08.01.2025
Information of all department leader during shop floor meeting QRQC	Joachim Gehl	08.01.2025	09.01.2025
Arrange sorting at customer site 152K pcs (So far 63.000 pcs. checked)	Linhua ZHU	08.01.2025	

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PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The grinding spindle was defective and stopped. After testing the frequency converter did not notice that the spindle was no longer rotating.The guiding rail was then emptied by the employee but he probably pushed the 2 parts to the already ground parts. That resulting that the two slots were not ground.

How was the defect reproduced ?

With blocked spindle.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

The hourly check could not detect the defective parts.
The machine did not stop automaticly after the grinding spindle stopped because the frequency converter did not send send a signal to stop the machine.
This was also not highlighted, because the worker was not fully aware of this defect.

D 6. Corrective actions and non detection

Corrective actions	Responsible	Date planned	Date closed
Replaced the defective spindle	Production	08.01.2025	08.01.2025
Check and adjust the frequency converter (FU)	G. Dohr	08.01.2025	08.01.2025
Highlight the alarm message and the red signal on the grinder	G. Dohr	09.01.2025	09.01.2025

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Responsible	Date planned	Date closed
Check during the next audit=> Signal is in place	Q-Team	09.01.2025	09.01.2025

D 8. Lessons learned

8.1 Do following documents require an update ?

	Responsible	Date planned	Date closed
P-FMEA	M. Ott	10.01.2025	10.01.2025
Control Plan			
Procedure			
Instructions			
Other			
Other			

8.2 Generalization of countermeasures

Generally, if one of the tools listed above is used. This is carefully checked and updated.

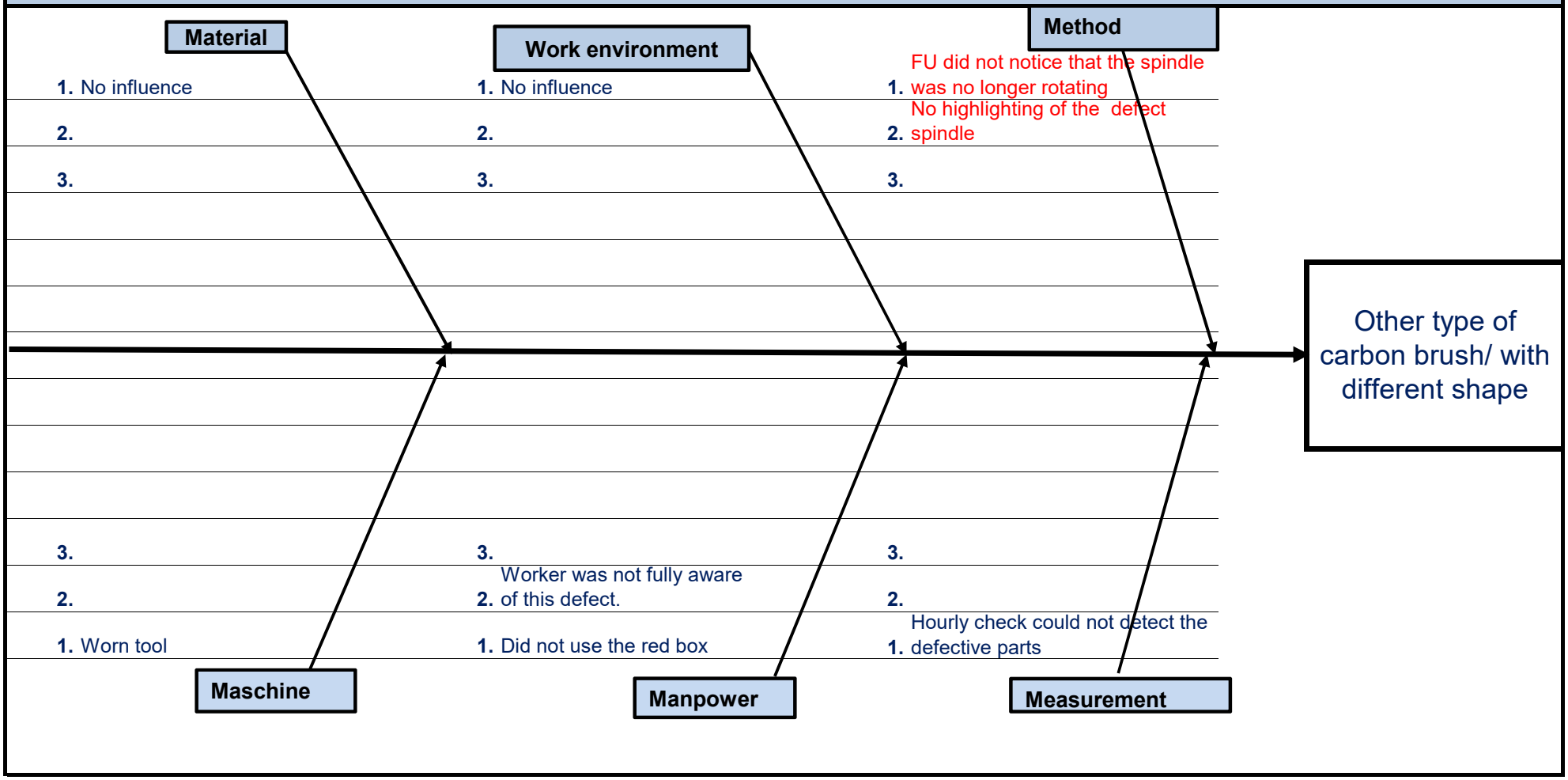
8.3 Internal audit (if required)

Date: 09.01.25 Results: OK

Comments

Signal is highlighted red when the machine stops with an error message

Ishikawa


Material
1. No influence
2.
3.
Work environment
1. No influence
2.
3.
Method
1. FU did not notice that the spindle was no longer rotating
2. No highlighting of the defect spindle
3.
Other type of carbon brush/ with different shape
3.
2.
1. Worn tool
Maschine
3.
2. Worker was not fully aware of this defect.
1. Did not use the red box
Manpower
3.
2.
1. Hourly check could not detect the defective parts
Measurement

5 Why's

Problem description : 1394320515 Surface damage and abnormal

OCCURENCE

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY?	5th ANSWER
	1	Why has the unground slot occured?	Because there where undground parts in the rail.	Why were there unground parts in the rail?	Because the spindle stopped.**TRC**	Why stopped the spindle?	Because the spindle was defect and the FU did not send a signal.	Why has the FU did not send a signal?	Because the tolerance was not properly setted. **MRC 1**	Why was the FU not properly setted?	Because the risk was wrongly considered in FMEA **MRC 2**

NON-DETECTION

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY ?	5th ANSWER
	1	Why was the unground slot not detected?	Because the FU did not send a signal to stop the machine.	Why has the machine not stopped?	Because the FU was not properly setted.	Why was the FU not properly setted?	Because the risk was wrongly considered. **MRC**				
	2	Why was the unground slot not detected?	Because the worker was not aware.	Why was the worker not aware?	Because the machine stopped without a message or	Why stopped the machine without a message or	Because the FU was not properly setted.	Why was the FU not properly setted?	Because the risk was wrongly considered.		

FMEA update

Funktion Zweck	Fehlerart	Besondere Merkmale	Fehler-Auswirkung	Fehler-Ursache	Fehler
Schleifen der Lauffläche	nicht geschliffene Lauffläche [Rekl Eckerle Rumänien 240702/ 240807]		Presshaut nicht entfernt (erhöhte Anlaufgeräusche; Anlaufverhalten nicht nach Vorgabe)	Elektrospindel defekt	Rote Ki

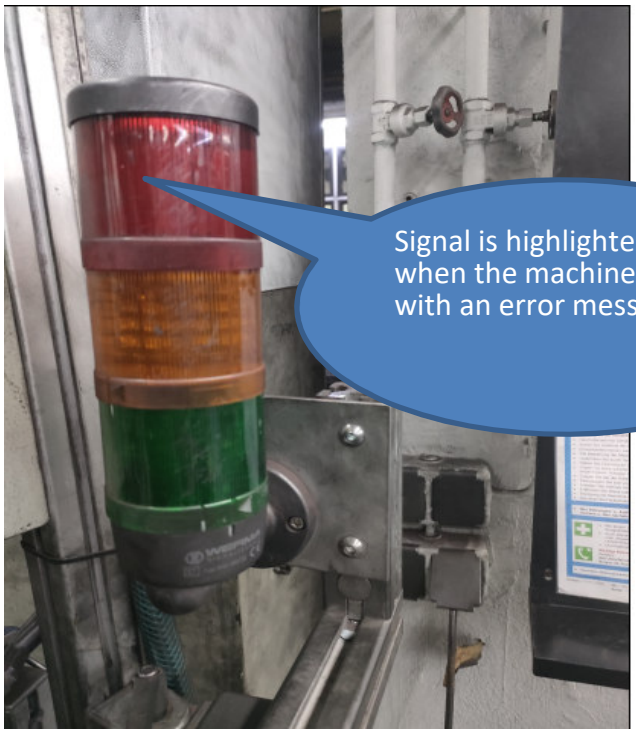
Not ground runing surface

Noises and starting behavior not acc. spec.

Spindle

Use of r

Foto of signal:



Signal is highlighted red when the machine stops with an error message

Maintenance tool:

Event & solution